Resilience & Disaster Recovery (DR)

Recovery Objectives

The architecture must stay available if an Availability Zone fails. It must also have a strategy for recovery documentation for a full Region outage. The project specifies 99.9% ingestion availability, a 15-minute telemetry RPO and 4-hour telemetry RTO, and a 24-hour warehouse RPO and 8-hour warehouse RTO.

RTO (Recovery Time Objective) is the maximum amount of time that is acceptable to restore a service after an outage. RPO (Recovery Point Objective) is the maximum amount that is acceptable for data loss, and it is measured in time.

Platform tier	RTO	RPO	Recovery approach
Plant telemetry ingestion	4 hours	15 minutes	Multi-AZ primary deployment, buffered plant gateways, cross-Region replication to a pilot-light environment
S3 data lake (bronze/silver/gold)	4 hours for critical ingestion; 8 hours for curated data	15 minutes for telemetry; up to 24 hours for curated data	S3 versioning and cross-Region replication; rebuild derived data from replicated raw data
Data warehouse (Redshift)	8 hours	24 hours	Automated snapshots copied to the DR Region; restore Redshift when needed
ETL and orchestration (Glue, Lambda, Step Functions)	8 hours	24 hours for scheduled jobs and workflow state	Infrastructure-as-code redeployment; retry or replay incomplete jobs
Dashboards and reporting (QuickSight)	8 hours	24 hours	Restore or recreate dashboards and reconnect them to the recovered warehouse
AI/ML services (SageMaker, Bedrock)	24 hours	24 hours for model artifacts and supporting data	Redeploy approved models and configurations; restore model artifacts and knowledge sources

The telemetry and warehouse objectives are project requirements. The other tier-level objectives are design targets that extend those requirements to supporting services.

1.2 Chosen DR Pattern: Multi-AZ Primary with Cross-Region Pilot Light

AeroForge uses a pilot-light strategy rather than a duplicate, fully-operational system in another region. The primary system runs on a plurality of availability zones within a single U.S. region. A second U.S. region maintains the minimum of facilities and data needed for the recovery of certain critical services.

The solution seeks to balance the recovery needs with the cost of a multi-region or full-duplication solution. A multi-site or full duplication solution would recover faster, but it would require the duplication of much more processing, compute, and storage resources even if there were no disasters.

Primary Region

- Deploy ingestion, processing, storage, and warehouse services across multiple Availability Zones.
- Use Kinesis Data Streams and Firehose for telemetry ingestion and delivery.
- Use S3 versioning and lifecycle policies for data protection.
- Use Redshift automated snapshots and AWS Backup where appropriate.
- Buffer telemetry at plant gateways so temporary network or cloud interruptions do not immediately result in data loss.

DR Region

- Establish the VPC, subnets, IAM roles, KMS keys, security controls, and required service configurations using infrastructure as code.
- Replicate critical S3 data to a second U.S. Region using S3 Cross-Region Replication (CRR).
- Maintain a recovery ingestion path. A replication process forwards telemetry to a DR-region stream, while plant gateways retain buffered events for replay if connectivity is interrupted.
- Copy Redshift snapshots to the DR Region on a scheduled basis.
- Keep Glue job definitions, Lambda code, Step Functions definitions, and other deployment artifacts available for redeployment.
- Restore Redshift compute and other higher-cost services only when a disaster is declared

Recovery sequence

1: Detect an outage by CloudWatch alarms and operational monitoring, and report it to the response team

2: Declare the disaster and authorize the recovery through the incident response system

3: Activate the DR-region ingestion path and restore the required network and security configuration

4: Replay the queued telemetry and recover the replicated data

5: Restore the Redshift warehouse from the latest available snapshot

6: Run any jobs left in the Glue queue to confirm data quality, record counts, totals, and so on

7: Restore the business intelligence dashboards and operational reports and notify the plant and finance stakeholders.

1.3 Cost Estimates for a DR Plan

The following preliminary planning estimates for incremental DR costs are not based on a complete AWS cost calculation. Actual costs will vary depending on the region, the number of copies, the duration of the copies, snapshot size, and the frequency of disaster recovery testing.

Cost component	3-plant pilot	12-plant environment
S3 cross-Region replication storage and requests	$15–$60/month	$50–$180/month
Cross-Region data transfer	$10–$40/month	$30–$120/month
Redshift snapshot storage and backup copies	$25–$100/month	$100–$350/month
Minimal DR infrastructure, monitoring, and testing	$100–$250/month	$200–$500/month
Estimated incremental DR total	**$150–$450/month**	**$380–$1,150/month**

These estimates assume the DR Region does not run a second full Redshift warehouse or duplicate all production compute continuously. They exclude temporary costs of operating the recovered environment during an actual disaster.

Cost-Control Measures

- Keep the DR Region in pilot-light mode and launch expensive compute only during recovery.
- Use S3 lifecycle policies to transition older replicated data to lower-cost storage classes when recovery requirements allow.
- Retain only the necessary number of Redshift snapshots and backup copies.
- Schedule recovery tests and shut down temporary test resources immediately afterward.

1.4 Export-Control and Data-Residency Considerations

The TALON military program requires special treatment. Its technical data must remain in U.S. Regions and be accessible only to authorized U.S. persons. Therefore, TALON data should replicate only between approved U.S. Regions, with separate IAM permissions, Lake Formation policies, and encryption keys. The DR process must preserve those restrictions; recovery must not make TALON data accessible to unauthorized users.

QRO data also requires care. Mexican-peso financial records and QRO company-code data should retain their currency and company-code distinctions during recovery. Mexican payroll and HR information should be excluded from cross-Region replication until its residency requirements have been assessed and approved.

A pilot-light strategy can meet the stated recovery objectives only if replication, gateway buffering, and restoration procedures are tested. AeroForge should conduct periodic recovery exercises to verify that telemetry can be recovered within 4 hours and that the warehouse can be restored within 8 hours.